

Software For ROBOTICS

Free Daily Utility Tools

- + **Office tools:** Apache OpenOffice (www.openoffice.org)
- + **Zippping software:** WinZip (www.winzip.com)
- + **Browser:** Mozilla Firefox (www.mozilla.org)
- + **PDF Reader:** Adobe Acrobat Reader DC (get.adobe.com)
- + **Media player:** PotPlayer (potplayer.daum.net)

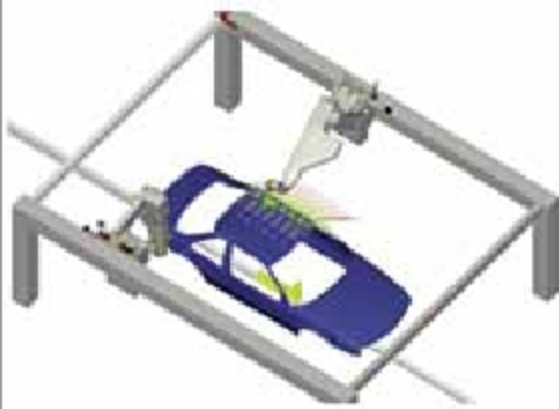
Ayushee Sharma

Roboguide

Visit: <https://www.fanucamerica.com>

Full version: Paid

Roboguide family of software packages allows users to create, program, and simulate a robotic workcell in 3D without the need for the Internet. For applications that require material handling, HandlingPRO comes with features like CAD to path programming and machine modelling. Others include PaintPRO for simplifying the painting process, PalletPRO for mainly developing workcell layouts, and WeldPRO for simulating the robotic arc welding process. This reduces the time and cost incurred in creating a physical prototype, besides avoiding the risks involved due to design problems after actual installation.



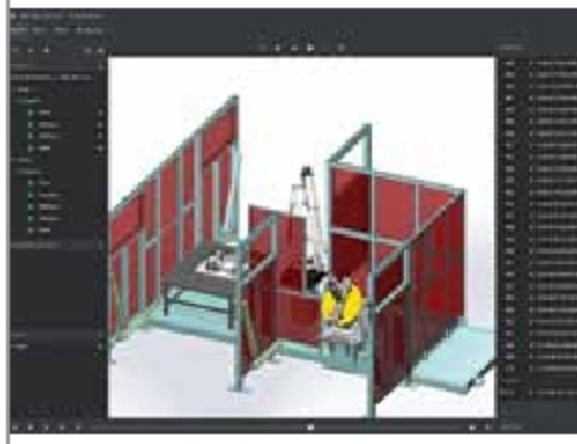
Workcell with simulation of robotic path playback

Robotmaster

Visit: <https://www.robotmaster.com/en>

Full version: Paid

Robotmaster offers a task-based robot programming platform for part manufacturing. This includes welding, cutting, drilling, fastening, and so on. The standalone tool is easy and intuitive, even if the users lack robotics or CAD/CAM expertise. All errors are visualised and mapped in a single screen and colour-coded by type. Click-and-drag interactive simulation/editing environment makes analysis simple and reduces time. Other features include offline programming, automated robot motion optimisation, accurate robot trajectories without the use of teaching points, terminology and control setting custom to the application, and the like.



Virtual Robotics Toolkit

Visit: <https://www.virtualrobotics-toolkit.com>

Full version: Paid

Virtual Robotics Toolkit by Cogmation Robotics is an educational software that allows users to design and program their digital robots. The development environment is made for use with Lego Mindstorms NXT and EV3 robots. Students can expand their knowledge and share their projects in the form of files without the need for additional resources. Testing for various environments and troubleshooting can be done without the requirement of physical space. Simulation lessons are also available for all skill levels.



CoderZ Learn

Visit: <https://gocoderz.com>; **Full version:** Paid; free trial

CoderZ Learn is a STEM education platform to teach students between 6th and 12th grade about programming of virtual 3D robots. The codes can be tested with real-time simulation. There is no need for prior experience or additional hardware and software. The online environment is designed in a fun and engaging way so that students can learn from anywhere in the world. Working on it also helps in understanding other STEM concepts like navigation, mechanics, and sensors.

